STAT-S 650 HOMEWORK 3

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Theory

1) Mathematically derive the auto covariance function for

$$Y(t) = w(t) + \theta_1 \cdot w(t-1) + \theta_2 \cdot w(t-2) + \theta_3 \cdot w(t-3)$$

lets assume $w(t)$ is white noise with

Mean = 0

Variance : σ_w^2

Covariance $\gamma_w(k) = \begin{cases} \sigma_w^2 & k=0 \\ 0 & k \neq 0 \end{cases}$

- Autocovariance function of the Moving average process

At lag k, the autocovariance is

$$\gamma_y(k) = \text{Cov}\left(Y(t), Y(t+k)\right).$$

$$Y(t) = \sum_{i=0}^{3} \theta_i w(t-i), \quad \theta_0 = 1$$

$$Y(t+k) = \sum_{j=0}^{3} \theta_j w(t+k-j)$$

$$\therefore \gamma_y(k) = \sum_{i=0}^{3} \sum_{j=0}^{3} \theta_i \theta_j \, \text{Cov}\left(w(t-j) \; w(t+k-j)\right).$$

White noise Cov is non zero only when the two indices match.

$$t - i = t + k - j \quad \Rightarrow \quad j = i + k.$$

Coupling $y(k)$ for each lag.

at lag 0

$$\gamma_y(0) = \sigma_w^2 \left(\theta_0^2 + \theta_1^2 + \theta_2^2 + \theta_3^2\right)$$

$$\Rightarrow \gamma_y(0) = \sigma_w^2 \left(1 + \theta_1^2 + \theta_2^2 + \theta_3^2\right).$$

at lag 1

$$\gamma_y(1) = \sigma_w^2 \left(\theta_1\theta_0 + \theta_2\theta_1 + \theta_3\theta_2\right)$$

$$\gamma_y(1) = \sigma_w^2 \left(\theta_1 + \theta_1\theta_2 + \theta_2\theta_3\right).$$

at lag 2

$$\gamma_y(2) = \sigma_w^2 \left(\theta_2\theta_0 + \theta_3\theta_1\right)$$

$$= \sigma_w^2 \left(\theta_2 + \theta_1\theta_3\right)$$

at lag 3

$$\gamma(3) = \sigma_w^2 \left(\theta_3\theta_0\right) =$$

$$= \sigma_w^2 \, \theta_3$$

8. The auto covariance function is

$$\gamma(k) = \begin{cases} \sigma_w^2 \left(1 + \theta_1^2 + \theta_2^2 + \theta_3^2\right) & k = 0 \\ \sigma_w^2 \left(\theta_1 + \theta_1\theta_2 + \theta_2\theta_3\right) & |k| = 1 \\ \sigma_w^2 \left(\theta_2 + \theta_1\theta_3\right) & |k| = 2 \\ \sigma_w^2 \theta_3 & |k| = 3 \\ 0 & |k| > 3 \end{cases}$$

2) Mathematically derive the cross cov function b/w $x(t-k)$ & $y(t)$ for the following model.

$$x(t) = \text{white noise}$$

$$y(t) = \alpha \cdot y(t-1) + x(t-3)$$

We need to find cross cov.

$$\gamma_{xy}(k) = \text{Cov}(x(t-k), y(t))$$

The model is $y(t) = \alpha y(t-1) + x(t-3)$

$$\Rightarrow \gamma_{xy}(k) = \text{Cov}(x(t-k), \alpha y(t-1) + x(t-3))$$

$$= \gamma_{xy}(k) = \alpha \,\text{Cov}(x(t-k), y(t-1)) + \text{Cov}(x(t-k), x(t-3))$$

$$\Rightarrow \underline{\text{Cov}(x(t-k), x(t-3))}$$

$$\text{Cov}(x(t-k), x(t-3)) = \begin{cases} \sigma_x^2 & t-k = t-3 \\ 0 & \text{otherwise} \end{cases}$$

For equality $k = 3$.

$$\text{Cov}(x(t-k), x(t-3)) = \begin{cases} \sigma_x^2 & k = 3 \\ 0 & k \neq 3 \end{cases}$$

$\Rightarrow$ $\underline{Cov\,(x(t-k),\ y(t-1))}$

$\Rightarrow$ $y(t-1) = \alpha y(t-2) + x(t-4)$

$x(t-4)$

$Cov\,(x(t-k),\ y(t-1)) = Cov\,(x(t-k),\ x(t-4))$

Non zero only when $k = 4$

$\Rightarrow$ $Cov\,(x(t-k),\ y(t-1)) = \begin{cases} \sigma_x^2 & k = 4 \\ 0 & k \neq 4. \end{cases}$

with α $\Rightarrow$ $\alpha\, Cov\,(x(t-k),\ y(t-1)) = \begin{cases} \alpha\sigma_x^2 & k = 4 \\ 0 & k \neq 4 \end{cases}$

$\therefore$ Final Cross Cov function:

$$\gamma_{xy}(k) = \begin{cases} \sigma_x^2 & k = 3 \\ \alpha\sigma_x^2 & k = 4 \\ 0 & \text{otherwise} \end{cases}$$